DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-200-03 — Replace VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1021/A2)

Area	A200 Vessels and column	**Task**	VR Valve replacement in kind
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	V-102 — V-102 vertical vessel 72" ID × 216" T/T — E 53'-6" N -25'-0" EL 100'-8-1/4"		
Equipment	V-102	**Lines**	4"-P300-P-1021
Isometrics	ISO-P-1021 (owner's drawings, attached)		
Permits	General work permit		

Reason

Outlet isolation valve stem packing leaking and seat passing.

Scope of work

1. Isolate per the isolation list, drain and vent 4"-P300-P-1021 between the isolations.
2. Unbolt and remove VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1021/A2) at E 53'-6" N -20'-8-3/16" EL 109'-0".
3. Inspect both flange faces (SW-NDE-5); dress or report damage.
4. Install a new VALVE, GATE, WEDGE, OS&Y 4" P300 of the same size, class and face-to-face (GATE4-P300), with new gaskets and studs, stem orientation as removed.
5. Tighten the studs to the stated preload in three passes, cross pattern (SW-WO-5).
6. Remove blinds, restore the line-up, leak test the two joints on start-up (SW-TEST-2).

Affected items: 4"-P300-P-1021/A2, 4"-P300-P-1021/A1, 4"-P300-P-1021/A4

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1021	4"-P300-P-1021	4"-P300-P-1021	4	2 error(s), 0 warning(s) — see below

ISO-P-1021 (PDF): ../isometrics-imported/ISO-P-1021.pdf — the sheets that hold the work follow the ITP.

Drawing	Sheet	Level	Finding
ISO-P-1021	1	error	sheet 1: CONT'D ON ISO-P-1021 SH 2 prints E 53'-6" N -13'-4 1/16" EL 109'-0" for E 53'-6" N -12'-7 15/16" EL 109'-0": the minus sign applies to the feet only; read as written the point is 8.12 in away (still on the centreline, at the wrong place along it)
ISO-P-1021	2	error	sheet 2: CONT'D FROM ISO-P-1021 SH 1 prints E 53'-6" N -13'-4 1/16" EL 109'-0" for E 53'-6" N -12'-7 15/16" EL 109'-0": the minus sign applies to the feet only; read as written the point is 8.12 in away (still on the centreline, at the wrong place along it)

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
V-102	equipment	V-102 vertical vessel 72" ID × 216" T/T		E 53'-6" N -25'-0" EL 100'-8-1/4"
4"-P300-P-1021/A2	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1021	E 53'-6" N -20'-8-3/16" EL 109'-0"
4"-P300-P-1021/A1	item	GASKET 4" P300 COMPRESSED SHEET	4"-P300-P-1021	E 53'-6" N -21'-3-7/8" EL 109'-0"
4"-P300-P-1021/A4	item	GASKET 4" P300 COMPRESSED SHEET	4"-P300-P-1021	E 53'-6" N -20'-1/2" EL 109'-0"
V-102-N2	nozzle	Nozzle V-102-N2 (outlet, 4" P300) on V-102		E 53'-6" N -21'-3-15/16" EL 109'-0"
4"-P300-P-1021/B2	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1021	E 44'-0" N 22'-6" EL 123'-10-13/16"
4"-P300-P-1021/B5	item	FLANGE WELD-NECK 4" P300 RAISED FACE	4"-P300-P-1021	E 44'-0" N 22'-6" EL 124'-9-5/16"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	equipment	V-102-N2	4"-P300-P-1021	Line end at nozzle V-102-N2: the equipment is out of service, drained and isolated under the shutdown isolation			
2	valve	4"-P300-P-1021/B2	4"-P300-P-1021	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1021/B2)	E 44'-0" N 22'-6" EL 123'-10-13/16"		
3	blind	4"-P300-P-1021/B5	4"-P300-P-1021	Fit a line blind between 4"-P300-P-1021/B2 and FLANGE WELD-NECK 4" P300 RAISED FACE (4"-P300-P-1021/B5), work side	E 44'-0" N 22'-6" EL 124'-6-9/16"		

Parts

Description	Qty	Replaces
VALVE, GATE, WEDGE, OS&Y 4" P300	1	4"-P300-P-1021/A2
GASKET 4" P300 COMPRESSED SHEET	1	4"-P300-P-1021/A1
STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	1	4"-P300-P-1021/A3
GASKET 4" P300 COMPRESSED SHEET	1	4"-P300-P-1021/A4
STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	1	4"-P300-P-1021/A6

Inspections

Id	Inspection	Criteria	Basis
WO-200-03-I1	Flange faces of both joints before the new gaskets go in	No radial scratch across the sealing face (SW-NDE-5)	SW-NDE-5

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
4"-P300-P-1021/A1	4 × 1" studs (4"-P300-P-1021/A3)	35,343	3	
4"-P300-P-1021/A4	4 × 1" studs (4"-P300-P-1021/A6)	35,343	3	

Test — Service leak test

Description	Both remade joints checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

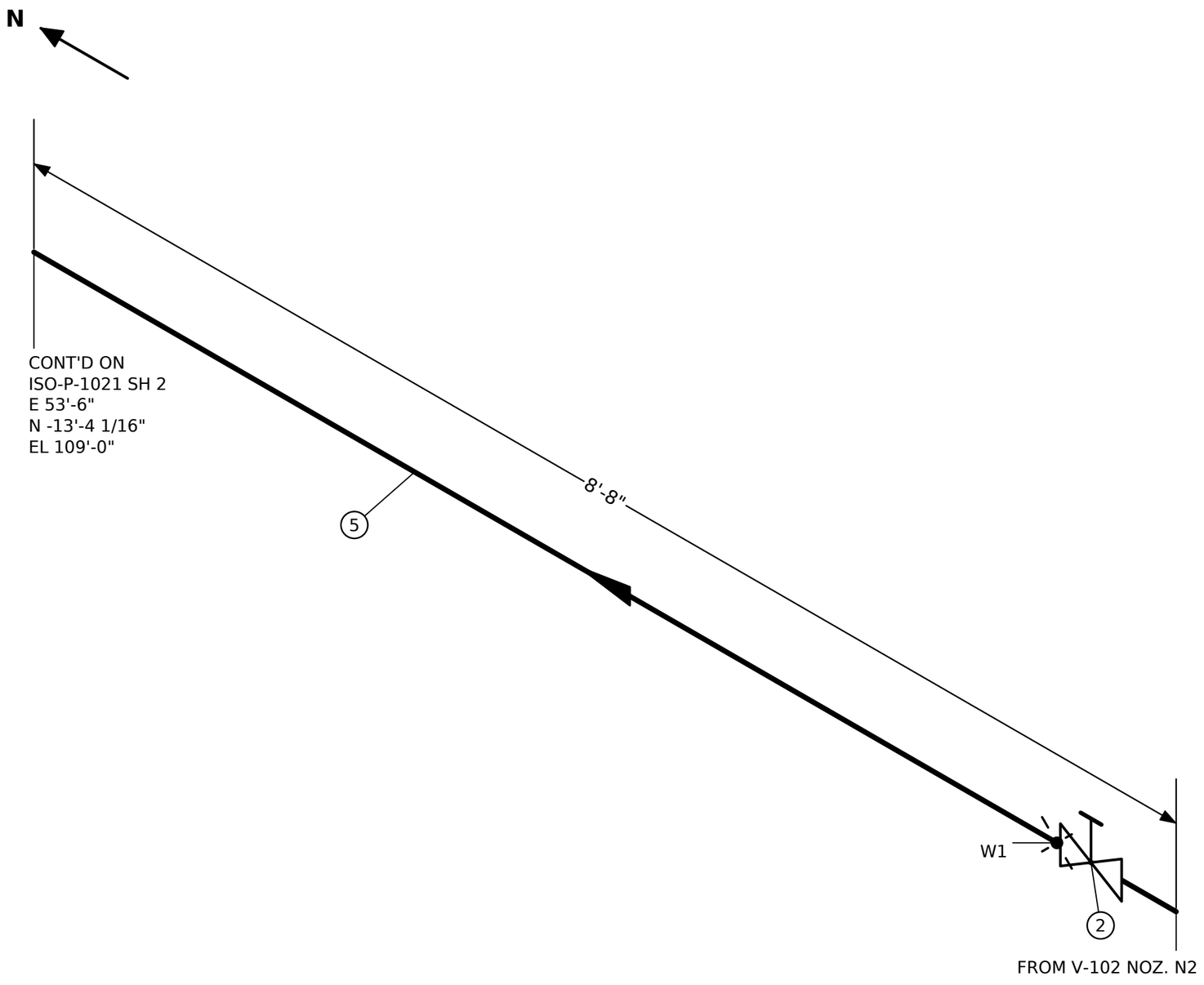
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Valve 4"-P300-P-1021/A2 removed; flange faces inspected	SW-NDE-5	No radial scratch across the sealing face	Flange face inspection record	W	W	—				
3	New valve verified: type, size, class, face-to-face and tag	GATE4-P300	Matches the parts list and the line class	Material receiving record	R	R	—				
4	New gaskets fitted, studs tightened to preload in three passes	SW-WO-5	Preload per stud as stated; three passes, cross pattern	Bolt-up record per joint	W	W	—				
5	Service leak test of the remade joints on start-up	SW-TEST-2	No leak at either joint	Leak test record	W	W	—				
6	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
7	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

Tag	Role	Drawing	Sheet	Weld label
4"-P300-P-1021/A2	affected	ISO-P-1021	1	
4"-P300-P-1021/A1	affected	ISO-P-1021	1	
4"-P300-P-1021/A4	affected	ISO-P-1021	1	
4"-P300-P-1021/B2	isolation: valve	ISO-P-1021	4	
4"-P300-P-1021/B5	isolation: blind	ISO-P-1021	4	



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	4"	GASKET 4" P300 COMPRESSED SHEET
2	1	4"	VALVE, GATE, WEDGE, OS&Y 4" P300
3	2	4"	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300
4	1	4"	FLANGE WELD-NECK 4" P300 RAISED FACE
5	6'-11 1/16"	4"	PIPE 4" P300 BEVELLED ENDS

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. FLANGE BOLT HOLES STRADDLE CENTRELINES
4. PAINT PER PROJECT SPECIFICATION

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

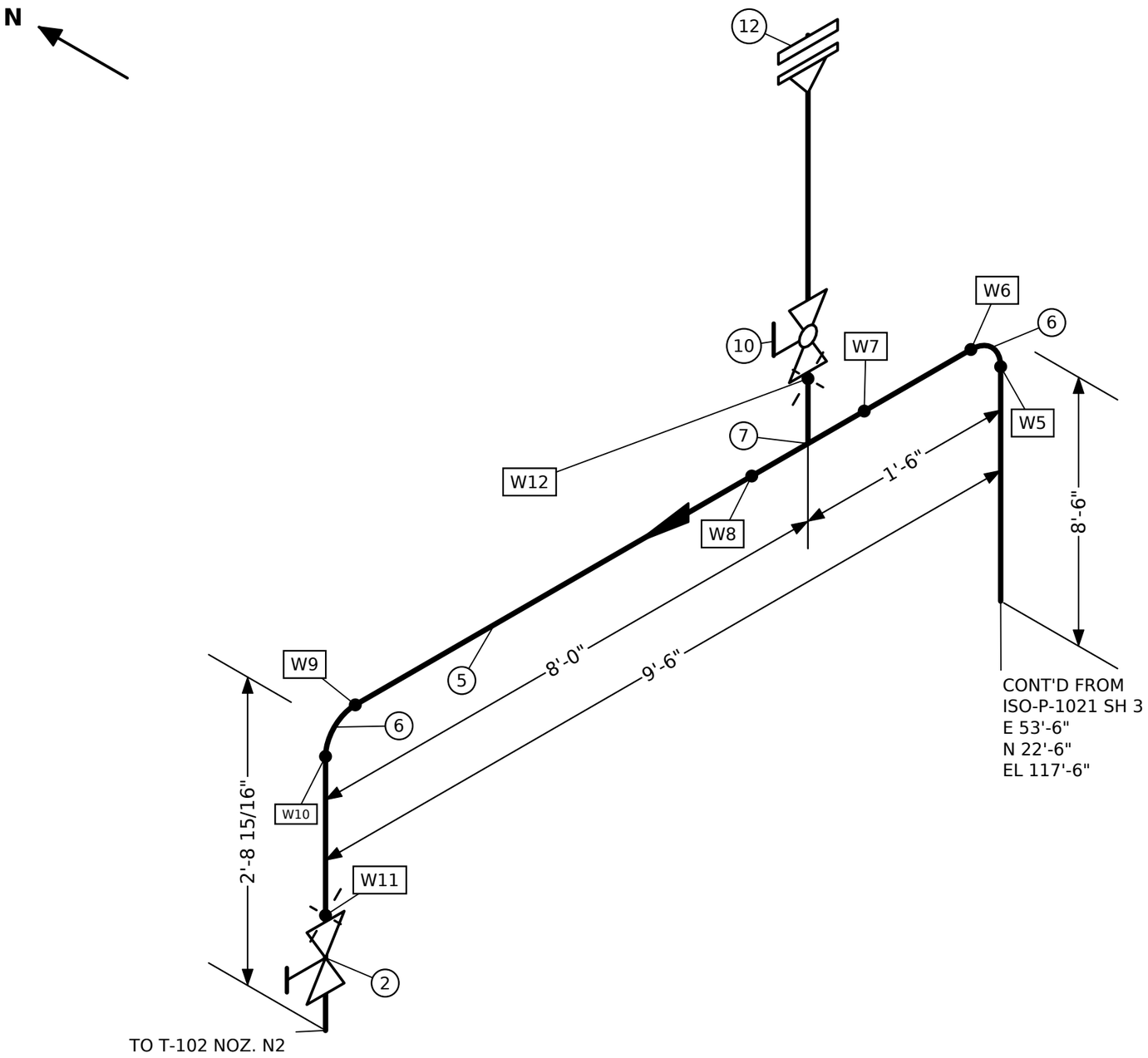
TEST PRESS.	INSULATION	PAINT
-	NONE	-

PWHT	NDE	FLUID
-	-	P

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
4"-P300-P-1021	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1021	1 OF 4	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	4"	GASKET 4" P300 COMPRESSED SHEET
2	1	4"	VALVE, GATE, WEDGE, OS&Y 4" P300
3	2	4"	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300
4	1	4"	FLANGE WELD-NECK 4" P300 RAISED FACE
5	16'-2 11/16"	4"	PIPE 4" P300 BEVELLED ENDS
6	2	4"	ELBOW 90° R = 5/3 OD 4" P300
7	1	4"x3/4"	TEE REDUCING 4" × 3/4" P300
8	1	3/4"	FLANGE WELD-NECK 3/4" P300 RAISED FACE
9	2	3/4"	GASKET 3/4" P300 COMPRESSED SHEET
10	1	3/4"	VALVE, BALL, FLOATING 3/4" P300
11	2	3/4"	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300
12	1	3/4"	FLANGE BLIND 3/4" P300 RAISED FACE

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	P

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
4"-P300-P-1021	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1021	4 OF 4	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS

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